

measured in a corresponding cluster in the electromagnetic calorimeter (ECAL), and the sum of all bremsstrahlung photons compatible with the electron trajectory. Muons are reconstructed from hits in the tracker and muon system. Charged hadrons are measured by a combination of tracker and the connected energy clusters in the ECAL and hadronic calorimeter (HCAL). Photons and neutral hadrons are reconstructed from energy clusters in the ECAL and a combination of ECAL and HCAL, respectively.

The primary pp interaction vertex is taken to be the vertex corresponding to the hardest scattering in the event, evaluated using tracking information alone, as described in Section 9.4.1 of Ref. [94]. In order to reduce effects from additional pp collisions in each event, we use pileup mitigation tools that act on the list and remove PF candidates that can be associated with a pileup vertex. The CMS Collaboration uses two algorithms for pileup mitigation. The charge-hadron subtraction (CHS) [95] technique removes charged hadrons that are associated with a pileup vertex by calculating the distance of closest approach of each track to the reconstructed primary vertices. The PU-per-particle identification (PUPPI) [96, 97] algorithm goes one step further and also acts on neutral PF candidates. Each PF candidate is assigned a weight between 0 and 1 that scales the four-momentum according to the probability of the particle to originate from a pileup interaction. The weight is calculated as a function of a variable defined by the energy deposits in the vicinity of the PF candidate. The PUPPI algorithm makes the additional pileup corrections to jets unnecessary, and has improved the performance and pileup stability of jet substructure tagging.

The modified list of PF candidates is subsequently used as input for jet clustering algorithms, such that hadronic decay products of the top quark can be identified with jets. In CMS, the anti- k_T [98] jet clustering algorithm is commonly used, as implemented in the FASTJET software package [99] using a distance parameter of $R = 0.4$. The missing transverse momentum vector $\vec{p}_T^{\text{miss}}$ is computed as the negative vector sum of the transverse momenta of all the PF candidates in an event, and its magnitude is denoted as p_T^{miss} [100]. The jet energy scale (JES) [101] is corrected for pileup effects, detector effects, and residual differences between data and simulation. The jet energy resolution (JER) [101] is smeared in simulated events in order to match the resolution observed in data. Both corrections are propagated to p_T^{miss} in each event.

Jets originating from b quarks are identified (tagged) with multivariate approaches that make use of global event, secondary vertex, displaced track, and jet constituent information [102].

2.3 Kinematic reconstruction of the $t\bar{t}$ system

The top quarks are investigated experimentally by measuring their decay products and their kinematic properties. In the all-jets decay channel, all decay products are reconstructed. In the dilepton channel, however, the two neutrinos from the W boson decay are not measured, thus leading to ambiguities in the reconstruction of neutrino momenta. The lepton+jets channel exhibits intermediate properties with only one neutrino in the final state, leading to fewer ambiguities. Several methods of kinematic reconstruction of $t\bar{t}$ pairs have been developed, which are described in the following.

2.3.1 Reconstruction in the lepton+jets and all-jets channels

In the lepton+jets and all-jets channels, kinematic fits [103, 104] are employed to check the compatibility of an event with the $t\bar{t}$ hypothesis and to improve the resolution of the reconstructed quantities. The fit parameters are the three-vectors of the momenta of the six decay products resulting in 18 unknowns. The following constraints are applied in the fit: the invariant masses of the two top quark candidates should be the same and the invariant masses of both W boson

candidates should be 80.4 GeV [1].

In the lepton+jets channel, the four-momenta of the lepton and the four highest- p_T (leading) jets, and $\vec{p}_T^{\text{miss}}$ are the inputs that are fed together with their resolutions to the fit algorithm [103]. With these input values, the fit has two degrees of freedom. In the all-jets channel, the momenta and resolutions of the six leading jets are the inputs to the fitter [104] resulting in a fit with three degrees of freedom. The kinematic fit then minimises $\chi^2 \equiv (\mathbf{x} - \mathbf{x}^m)^T G (\mathbf{x} - \mathbf{x}^m)$, where $\mathbf{x}^m$ and $\mathbf{x}$ are the vectors of the measured and fitted momenta, respectively, and G is the inverse covariance matrix, which is constructed from the uncertainties in the measured momenta. The above-mentioned constraints are added to the minimisation procedure with Lagrange multipliers.

The fit is performed for all possible assignments of the jets to the decay products. To reduce combinatorics, exactly two of the selected leading jets are required to be identified as originating from a b quark (b tagged). In the lepton+jets channel, the two b -tagged jets are candidates for the b quarks in the $t\bar{t}$ hypothesis, while the two jets that are not b tagged serve as candidates for the light quarks from the hadronically decaying W boson. In addition, there are two solutions for the start value of the longitudinal component of the neutrino momentum per parton-jet assignment. Hence, the fit is performed for four different permutations per event. In the all-jets channel, the two b -tagged jets are the candidates for the b quarks and the four jets that are not b tagged serve as candidates for the light quarks from the hadronically decaying W bosons. Hence, the fit is performed for six different permutations.

The χ^2 probability P_{gof} of the kinematic fits is used to rank the permutations, since the permutations with wrongly assigned jets typically have very low P_{gof} values. For simulated $t\bar{t}$ events, the parton-jet assignments can be classified as correct, wrong, and unmatched permutations. In the first case, all quarks from the $t\bar{t}$ decay are matched within a distance of $\Delta R = \sqrt{(\Delta\eta)^2 + (\Delta\phi)^2} < 0.3$, where ϕ is the azimuthal angle and η is the pseudorapidity, to a selected jet and assigned with the correct flavour assumption to the correct top quark. If all quarks are matched to a selected jet, but the wrong permutation is chosen, it is labelled ‘wrong’, while ‘unmatched’ indicates that not all quarks are matched unambiguously to a selected jet.

Due to the constraints, the kinematic fits improve the resolution of the reconstructed mass of the top quark candidates. The resolution of the reconstructed mass of the top quark with and without applying the kinematic fit is presented in Fig. 3 for the lepton+jets (multiple permutations) and all-jets channels (permutation with lowest χ^2). In the all-jets channel, only the permutation with the lowest χ^2 in each event is considered for further analysis. The resolution σ^{peak} is extracted by fitting a Gaussian distribution within the range $-40 < m_t^{\text{rec}} - m_t^{\text{gen}} < +40$ GeV. Without a kinematic fit, the resolution of the reconstructed top quark mass is relatively poor in the case of the lepton+jets channel, while the peak is hardly discernible at all in the all-jets channel. In both $t\bar{t}$ decay channels, the kinematic fit improves the resolution using either all jet-parton permutations or the one with the lowest χ^2 . Finally, a cut on $P_{\text{gof}} > 0.2$ (0.1) is used in the lepton+jets (all-jets) channel, which matches the resolution of the case where only correct permutations are considered with their pre-fit momenta. The selection efficiency of the P_{gof} cut is 27.4 (5.3)% in the lepton+jets (all-jets) channel. Besides the mass, the kinematic fits can also improve the reconstruction of other kinematic variables of the $t\bar{t}$ system, such as its invariant mass $m_{t\bar{t}}$. The bias and resolution of the reconstructed $m_{t\bar{t}}^{\text{rec}}$ with regard to the generated $m_{t\bar{t}}^{\text{gen}}$ is shown for the lepton+jets channel in Fig. 4 and for the all-jets channel in Fig. 5. The resolution is defined as the root-mean-square (RMS) of the difference between the reconstructed and the generated parton-level quantity, and the bias as its mean. The kinematic fit with a P_{gof} cutoff improves the resolution and is almost free of bias over the examined range in $m_{t\bar{t}}^{\text{gen}}$.

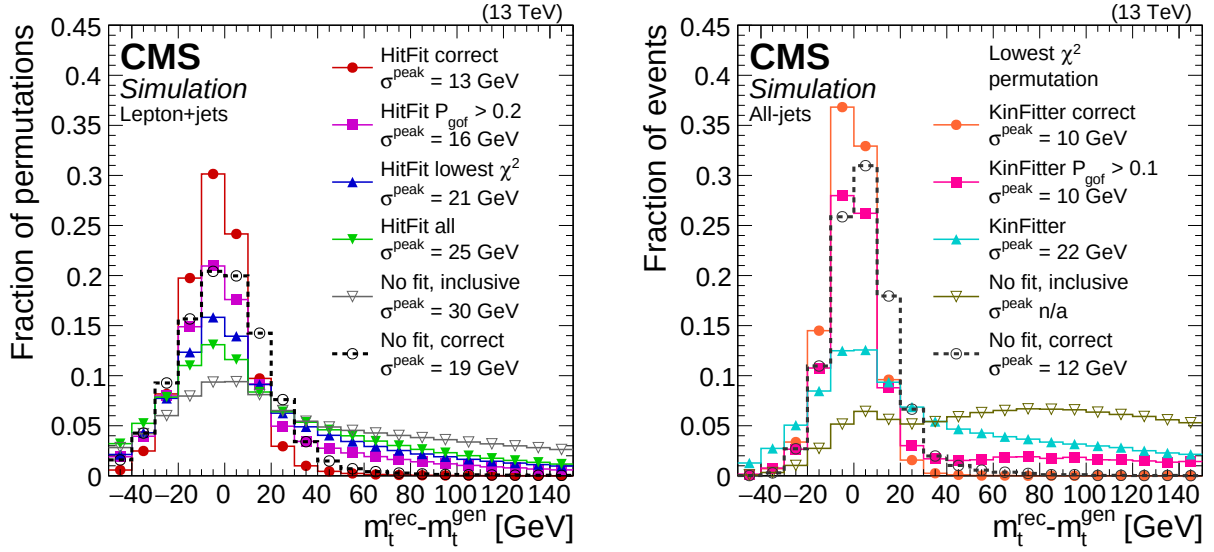


Figure 3: Reconstructed top quark mass resolution with and without the HITFIT/KINFITTER kinematic reconstruction in the lepton+jets (left) and all-jets (right) channels. Multiple reconstruction options with and without kinematic fit are represented by lines of different colour, and “correct” denotes the correct parton-jet assignments as discussed in the text. The HITFIT/KINFITTER reconstruction with a cutoff on P_{gof} is used for measuring the top quark mass [61, 62].

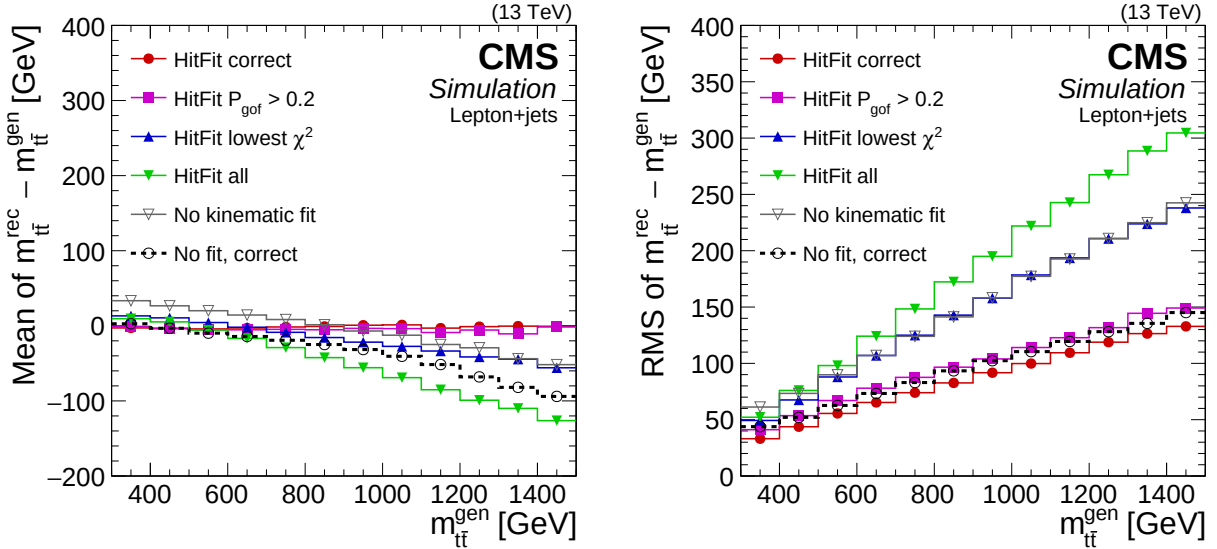


Figure 4: Reconstructed $t\bar{t}$ mass bias (left) and resolution (right) with and without the HITFIT kinematic reconstruction in the lepton+jets channel, as functions of the $t\bar{t}$ invariant mass at generator level. Multiple reconstruction options with and without kinematic fit are represented by lines of different colour, and “correct” denotes the correct parton-jet assignments as discussed in the text. The HITFIT reconstruction with a cutoff on P_{gof} is used for measuring the top quark mass [61].

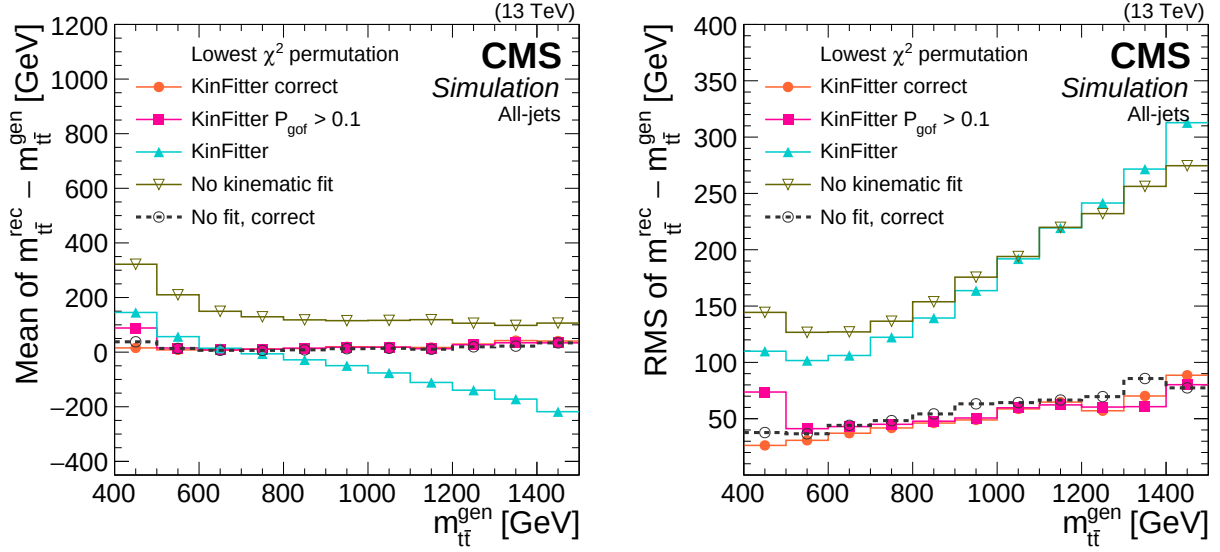


Figure 5: Reconstructed $t\bar{t}$ mass bias (left) and resolution (right) with and without the KINFITTER kinematic reconstruction in the all-jet channel, as functions of the $t\bar{t}$ invariant mass at generator level. Multiple reconstruction options with and without kinematic fit are represented by lines of different colour, and “correct” denotes the correct parton-jet assignments as discussed in the text. The KINFITTER reconstruction with a cutoff on P_{gof} is used for measuring the top quark mass [62].

2.3.2 Reconstruction in the dilepton channel

In contrast to the lepton+jets channel, direct measurements of m_t in the dilepton channel are challenging because of the ambiguity due to the two neutrinos in the final state, reconstructed as $\vec{p}_T^{\text{miss}}$. Therefore, the dilepton $t\bar{t}$ events are mostly used for extraction of the Lagrangian m_t through comparisons of the measurements of inclusive or differential $t\bar{t}$ cross sections [64, 65, 69] to the theoretical predictions, as explained in Section 4. In this case, the reconstruction method aims to obtain good resolution of the observable of interest and a high reconstruction efficiency.

For the $t\bar{t}$ reconstruction in the dilepton channel, several methods have been developed, with the primary task of obtaining solutions for the two unknown neutrino momenta. Depending on the observable of interest, either the individual top quark and antiquark, e.g. in the measurement of single-particle kinematics, or only the $t\bar{t}$ system, e.g. in the measurement of $m_{t\bar{t}}$ distribution, are reconstructed.

The **full kinematic reconstruction** (FKR) of the $t\bar{t}$ pair is based on the algebraic approach suggested in Ref. [105]. A system of kinematic equations describing the $t\bar{t}$ system is solved using the four-momenta of the six final-state particles, i.e. two leptons, two b jets, and the two neutrinos. It is assumed that the total measured missing transverse momentum is due to the two neutrinos and can be decomposed as follows:

$$p_x^{\text{miss}} = p_{x,\nu} + p_{x,\bar{\nu}}, \quad p_y^{\text{miss}} = p_{y,\nu} + p_{y,\bar{\nu}}. \quad (1)$$

The invariant mass of the lepton and the neutrino from the same top quark should correspond to the mass of the W boson, resulting in the following equations:

$$m_{W+}^2 = (E_{\ell+} + E_{\nu})^2 - (p_{x,\ell+} + p_{x,\nu})^2 - (p_{y,\ell+} + p_{y,\nu})^2 - (p_{z,\ell+} + p_{z,\nu})^2, \quad (2)$$

$$m_{W-}^2 = (E_{\ell-} + E_{\bar{\nu}})^2 - (p_{x,\ell-} + p_{x,\bar{\nu}})^2 - (p_{y,\ell-} + p_{y,\bar{\nu}})^2 - (p_{z,\ell-} + p_{z,\bar{\nu}})^2. \quad (3)$$

Finally, the masses of the top quark and antiquark are given, respectively, by:

$$m_t^2 = (E_{\ell^+} + E_\nu + E_b)^2 - (p_{x,\ell^+} + p_{x,\nu} + p_{x,b})^2 - (p_{y,\ell^+} + p_{y,\nu} + p_{y,b})^2 - (p_{z,\ell^+} + p_{z,\nu} + p_{z,b})^2, \quad (4)$$

$$m_{\bar{t}}^2 = (E_{\ell^-} + E_{\bar{\nu}} + E_{\bar{b}})^2 - (p_{x,\ell^-} + p_{x,\bar{\nu}} + p_{x,\bar{b}})^2 - (p_{y,\ell^-} + p_{y,\bar{\nu}} + p_{y,\bar{b}})^2 - (p_{z,\ell^-} + p_{z,\bar{\nu}} + p_{z,\bar{b}})^2. \quad (5)$$

The masses of the b quarks are set to the values used in the simulation, while lepton masses are assumed to be negligible. The masses of the top quark and of the W boson need to be fixed in order to solve the system of equations (1)–(5). For analyses where the choice does not directly affect the result of the measurement, they are typically fixed to the default values of $m_t = 172.5 \text{ GeV}$ and $m_W = 80.4 \text{ GeV}$. The equation system can then be solved analytically with a maximum four-fold ambiguity. Selected is the solution which yields the minimum invariant mass of the $t\bar{t}$ system, as it was shown that this choice provides the best solution in most cases. In analyses that target direct reconstruction of m_t in the dilepton channel, a dedicated method [47, 49, 53] is used that tests different m_t hypotheses. In contrast, in differential measurements of the $t\bar{t}$ cross section, the dependence on the choice of m_t in the reconstruction is usually estimated by varying the top quark mass assumption in the MC simulation.

To capture the effects of the finite detector resolution, the kinematic reconstruction is repeated 100 times, each time randomly smearing the measured energies and directions of the reconstructed leptons and jets within their resolutions. This smearing procedure recovers events that initially yielded no solution because of limited experimental resolution. Further, in the same smearing procedure, the mass of the W boson is varied according to a relativistic Breit–Wigner function, estimated using the generator-level W boson mass distribution. For each solution, a weight is calculated based on the expected true spectrum of the invariant mass of a lepton and a b jet ($m_{\ell b}$) stemming from the decay of a top quark and taking the product of the two weights for the top quark and antiquark decay chains: $w = w_{m_{\ell b}} w_{m_{\ell \bar{b}}}$. The final three-momenta of the top quarks j and k are then determined as a weighted average over all smeared solutions summing over all 100 kinematic reconstructions:

$$\langle \vec{p}_t^{k,j} \rangle = \frac{1}{w_s} \sum_{i=1}^{100} w_i \vec{p}_{t,i}^{k,j}, \quad \text{with} \quad w_s = \sum_{i=1}^{100} w_i. \quad (6)$$

All possible lepton-jet combinations in the event that satisfy the requirement for the invariant mass of the lepton and jet $m_{\ell b} < 180 \text{ GeV}$ are considered. Combinations are ranked, based on the presence of b-tagged jets in the assignments, i.e. a combination with both leptons assigned to b-tagged jets is preferred over those with one or zero b-tagged jet. Among assignments with an equal number of b-tagged jets, the one with the highest sum of weights is chosen. Events with no solution after smearing are discarded. The four-momentum vector of the top quark is determined by its energy, which is calculated from $\langle \vec{p}_t \rangle$, and the top quark mass of 172.5 GeV . The kinematic properties of the top antiquark are determined analogously. The efficiency of the kinematic reconstruction, defined as the number of events where a solution is found divided by the total number of selected $t\bar{t}$ events, is studied in data and simulation, and consistent results of about 90% are found in analyses at $\sqrt{s} = 13 \text{ TeV}$.

The value of the invariant mass $m_{t\bar{t}}$ of the $t\bar{t}$ pair obtained using FKR is highly sensitive to the predefined value of the top quark mass used as a kinematic constraint. However, the objective of the analyses described in this paper is the extraction of m_t , in some cases exploiting the $m_{t\bar{t}}$ distribution or related observables. For such cases, the **loose kinematic reconstruction (LKR)**

was developed [64], where the value of the top quark mass is not constrained. In this algorithm, the $\nu\bar{\nu}$ system is reconstructed, rather than the individual ν and $\bar{\nu}$. As a consequence, only the $t\bar{t}$ system can be reconstructed in LKR, but not the individual top quark and antiquark. As in FKR, all possible lepton-jet combinations in the event that satisfy the requirement for the invariant mass of the lepton and jet $m_{\ell b} < 180 \text{ GeV}$ are considered. Combinations are ranked, based on the presence of b-tagged jets in the assignments, but from all the combinations with an equal number of the b-tagged jets, the ones with the highest p_T jets are chosen. The kinematic variables of the $\nu\bar{\nu}$ system are derived as follows:

1. the transverse momentum $\vec{p}_T$ of the $\nu\bar{\nu}$ system is set equal to $\vec{p}_T^{\text{miss}}$;
2. the $\nu\bar{\nu}$ longitudinal momentum $p_{z,\nu\bar{\nu}}$ is set to that of the lepton pair, $p_{z,\nu\bar{\nu}} = p_{z,\ell\bar{\ell}}$, for $p_{T,\nu\bar{\nu}} < E_{\ell\bar{\ell}}$, and to zero otherwise;
3. the energy of the $\nu\bar{\nu}$ system $E_{\nu\bar{\nu}}$ is defined as $E_{\nu\bar{\nu}} = E_{\ell\bar{\ell}}$ for $p_{\nu\bar{\nu}} < E_{\ell\bar{\ell}}$, and $E_{\nu\bar{\nu}} = p_{\ell\bar{\ell}}$ otherwise, ensuring that $m_{\nu\bar{\nu}} \geq 0$;
4. the four-momentum sum of $\ell\bar{\ell}\nu\bar{\nu}$ is calculated;
5. for $m_{\ell\bar{\ell}\nu\bar{\nu}} < 2m_W = 2 \times 80.4 \text{ GeV}$, the mass component of the four-momentum of $\ell\bar{\ell}\nu\bar{\nu}$ is set to $2m_W$, ensuring that $m_{W+W-} \geq 2m_W$;
6. the four-momentum of the $t\bar{t}$ system is calculated by using the four-momenta of the $\ell\bar{\ell}\nu\bar{\nu}$ system and of the two b jets as $\ell\bar{\ell}\nu\bar{\nu}+b\bar{b}$.

The additional constraints that are applied on the invariant mass of the neutrino pair, $m_{\nu\bar{\nu}} \geq 0$ (item 3) and on the invariant mass of the W bosons, $m_{W+W-} \geq 2m_W$ (item 5) have only minor effects on the performance of the reconstruction. The method yields similar $t\bar{t}$ kinematic resolutions and reconstruction efficiency as for the FKR method. In the CMS analysis [64], the LKR was exclusively used to measure triple-differential $t\bar{t}$ cross sections as functions of the invariant mass and rapidity of the $t\bar{t}$ system, and the additional-jet multiplicity.

For the presented performance studies, the POWHEG+PYTHIA8 [106–110] $t\bar{t}$ simulated samples are used, which are explained in detail in Section 2.4. The reconstruction efficiency for both methods is shown in Fig. 6 as a function of the reconstructed $t\bar{t}$ kinematic variables $m_{t\bar{t}}$, $p_{T,t\bar{t}}$, and $y_{t\bar{t}}$. An event is considered as reconstructed if the reconstruction method yields at least one solution as described above. The overall efficiency for the LKR is about 4% higher than for the FKR, and shows the same kinematic properties. The maximum efficiency is achieved for low $m_{t\bar{t}}$, central $y_{t\bar{t}}$, and low $p_{T,t\bar{t}}$. The efficiency drops rapidly with increasing $p_{T,t\bar{t}}$ as the leptons and jets become less separated. For Lorentz-boosted configurations with $p_{T,t\bar{t}} > 700 \text{ GeV}$, the reconstruction fails in 20% of the cases.

The resolution and bias for both algorithms are shown in Figs. 7 and 8, respectively, as functions of the same three observables at the generator level. As described above for the lepton+jets decay channel, the resolution is defined as the RMS of the difference between the reconstructed and the parton-level quantity, and the bias as its mean. As in the case of the efficiencies, the LKR shows better performance. Its bias is often closer to zero in the low- $m_{t\bar{t}}$ regime, but becomes larger than in the case of the FKR for very large values of $m_{t\bar{t}}$. The LKR shows better resolution over the whole spectra, but it should be noted that the resolution definition is sensitive to outliers, e.g. in the tails of the distribution, affecting the performance of the FKR, e.g. in the low- $m_{t\bar{t}}$ region. For probing m_t in the dilepton channel, the resolution at low $m_{t\bar{t}}$, close to the

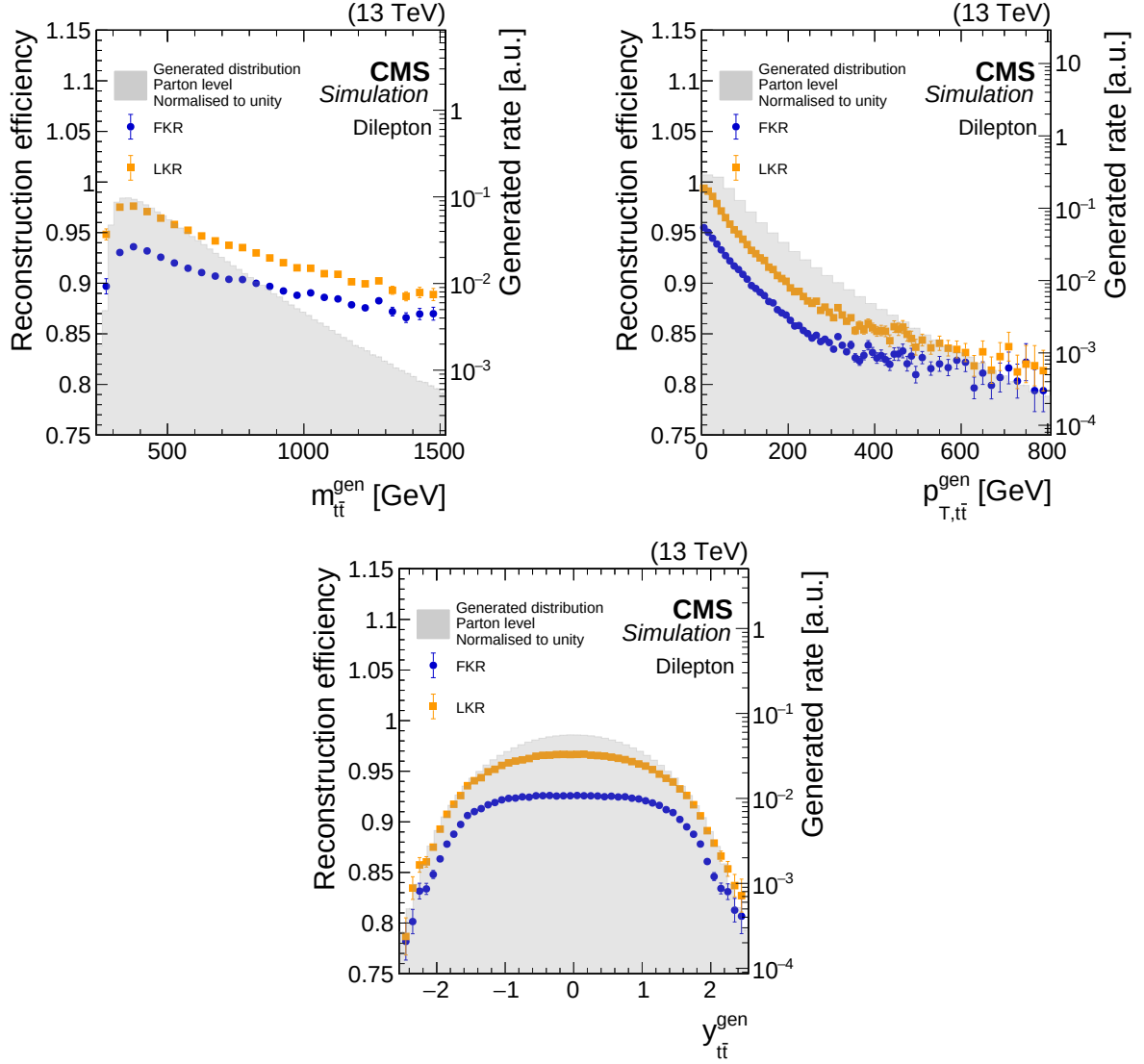


Figure 6: The reconstruction efficiencies for the full kinematic reconstruction (FKR, blue circles) and loose kinematic reconstruction (LKR, orange squares) are shown as functions of the invariant mass, transverse momentum, and rapidity of the reconstructed $t\bar{t}$ system. The averaged efficiencies are 92 (96)% for the FKR (LKR). The corresponding parton-generator-level distributions, normalised to unit area, for $t\bar{t}$ production are represented by the grey shaded areas, shown on the logarithmic scale (right y axis). The POWHEG+PYTHIA8 $t\bar{t}$ simulated samples are used.

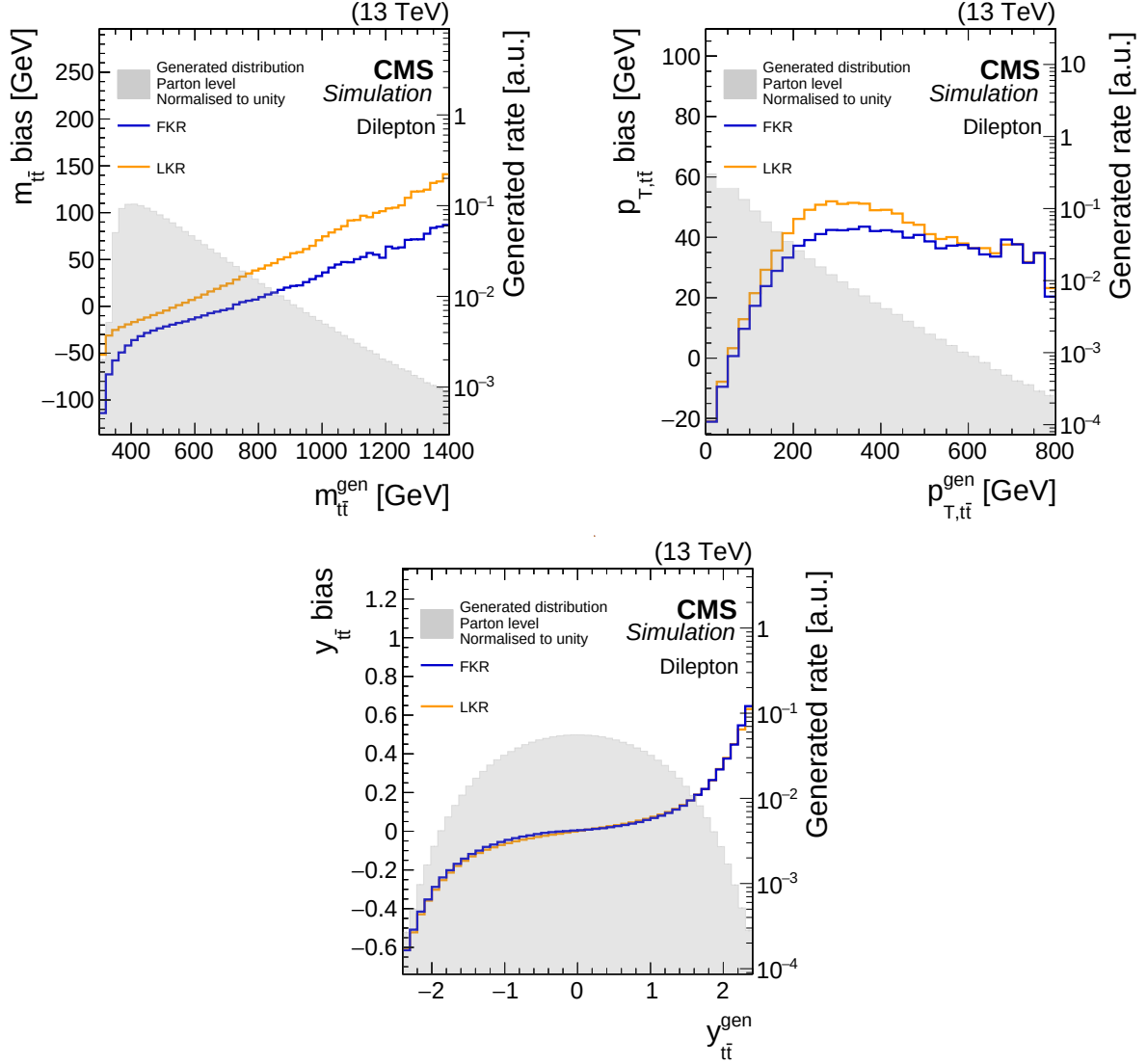


Figure 7: The biases (solid lines), as defined in the text, for the full kinematic reconstruction (FKR, blue) and loose kinematic reconstruction (LKR, orange) are shown for the invariant mass, transverse momentum, and rapidity of the $t\bar{t}$ system, as a function of the same variables at the generator level. The corresponding parton-generator-level distributions, normalised to unit area, for $t\bar{t}$ production are represented by the grey shaded areas, shown on the logarithmic scale (right y axis). The POWHEG+PYTHIA8 $t\bar{t}$ simulated samples are used.

production threshold, is of key importance. The resolution is about 100–150 GeV, which defines the minimal bin width in the differential $m_{t\bar{t}}$ measurement.

Since the FKR and LKR methods are developed to be agnostic to additional radiation for $t\bar{t}$ production, a multivariate method was developed in CMS [69] to optimise the resolution for an observable related to the invariant mass of the $t\bar{t}$ +jet system, denoted as ρ , which is defined for $t\bar{t}$ events with at least one additional jet:

$$\rho = \frac{340 \text{ GeV}}{m_{t\bar{t}+\text{jet}}}. \quad (7)$$

In the definition of ρ , the leading jet is considered and $m_{t\bar{t}+\text{jet}}$ is the invariant mass of the $t\bar{t}$ +jet system. This observable shows a large m_t sensitivity and is measured in a CMS analysis [69]

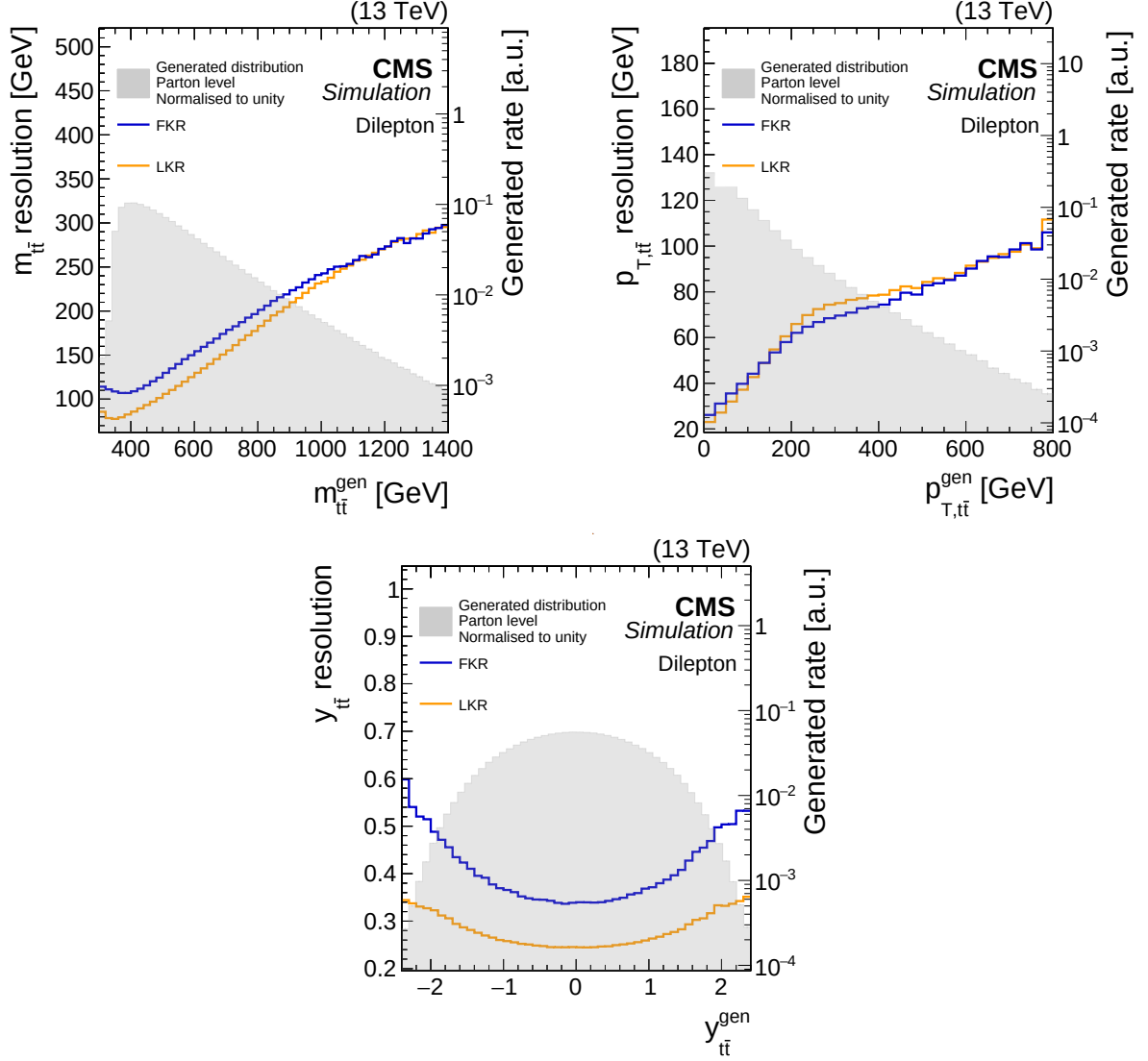


Figure 8: The resolutions (solid lines), as defined in the text, for the full kinematic reconstruction (FKR, blue) and loose kinematic reconstruction (LKR, orange) are shown as functions of the invariant mass, transverse momentum, and rapidity of the $t\bar{t}$ system at the generator level. The corresponding parton-generator-level distributions, normalised to unit area, for $t\bar{t}$ production are represented by the grey shaded areas, shown on the logarithmic scale (right y axis). The POWHEG+PYTHIA8 $t\bar{t}$ simulated samples are used.

described in Section 4.5 to extract m_t^{pole} . The result of the measurement is independent of the choice of the scaling constant in the numerator, which is introduced to define ρ dimensionless, and is on the order of two times m_t . Set up as a regressional neural network (NN), a fully connected feed-forward NN is trained. The benefit of using a regression NN is the maximised reconstruction efficiency, increasing the acceptance of the measurement, as it yields a solution for every event. The NN uses a set of low-level inputs, e.g. particle four-momenta, and high-level input variables, such as geometric and kinematic properties of the systems of the final-state objects. Starting from a set of 100 variables, the ten variables with the highest impact on the output of the NN are selected. These also include solutions of the LKR and FKR algorithms. Simulated events are used for the training of the regression NN if they contain at least three reconstructed jets with $p_T > 30 \text{ GeV}$ and $|\eta| < 2.4$. The ten input variables, ordered by their impact, used for the regression NN are:

- the calculation for ρ using the LKR;
- the calculation for ρ using the FKR;
- the invariant mass of the dilepton and subleading jet system;
- the invariant mass of the leading lepton and subleading jet system;
- the p_T of the subleading lepton;
- the invariant mass of the dilepton system;
- the invariant mass of the subleading lepton and subleading jet system;
- the invariant mass of the subleading lepton and leading jet system;
- the invariant mass of the dilepton and leading jet system;
- p_T^{miss} .

The training is performed using an independent data set, which is produced with the MADGRAPH5_aMC@NLO [111] event generator at NLO accuracy, interfaced with PYTHIA8 [110]. Afterwards, the resulting performance is also evaluated using the POWHEG+PYTHIA8 simulation, and is checked for possible overtraining.

The performance of the NN regression is shown in Fig. 9. The left plot shows the correlation between the parton-level value (ρ_{gen}) and the reconstructed value (ρ_{reco}). The correlation coefficient for the regression is 0.87, compared to 0.78 (0.84) for the loose (full) kinematic reconstruction.

The resolution of the regression NN is compared to that of the FKR and LKR in Fig. 9 (right). The resolution is defined as the RMS of the difference between the true value ρ_{gen} at parton level and the reconstructed value ρ_{reco} of the regression NN in a given ρ_{gen} bin, divided by $1 + \langle \rho_{\text{gen}} - \rho_{\text{reco}} \rangle$ to account for the bias in the reconstruction and to evaluate the response corrected resolution. The advantage of the multivariate method is the final resolution ranging between 0.05 and 0.08 in the full spectrum, which is an improvement by as much as a factor of two with respect to earlier approaches. The most significant improvement is achieved for the values of ρ_{gen} close to unity. Since this kinematic regime corresponds to small values for the invariant mass of the $t\bar{t}$ +jet system, it is the most sensitive region for the m_t measurement. An additional advantage is the 10–15% higher reconstruction efficiency since the described method is 100% efficient.

2.4 Monte Carlo simulations and modelling uncertainties

Physics generator configurations for top quark mass measurements Proton-proton collisions are modelled and studied using MC event generators, which split the prediction into